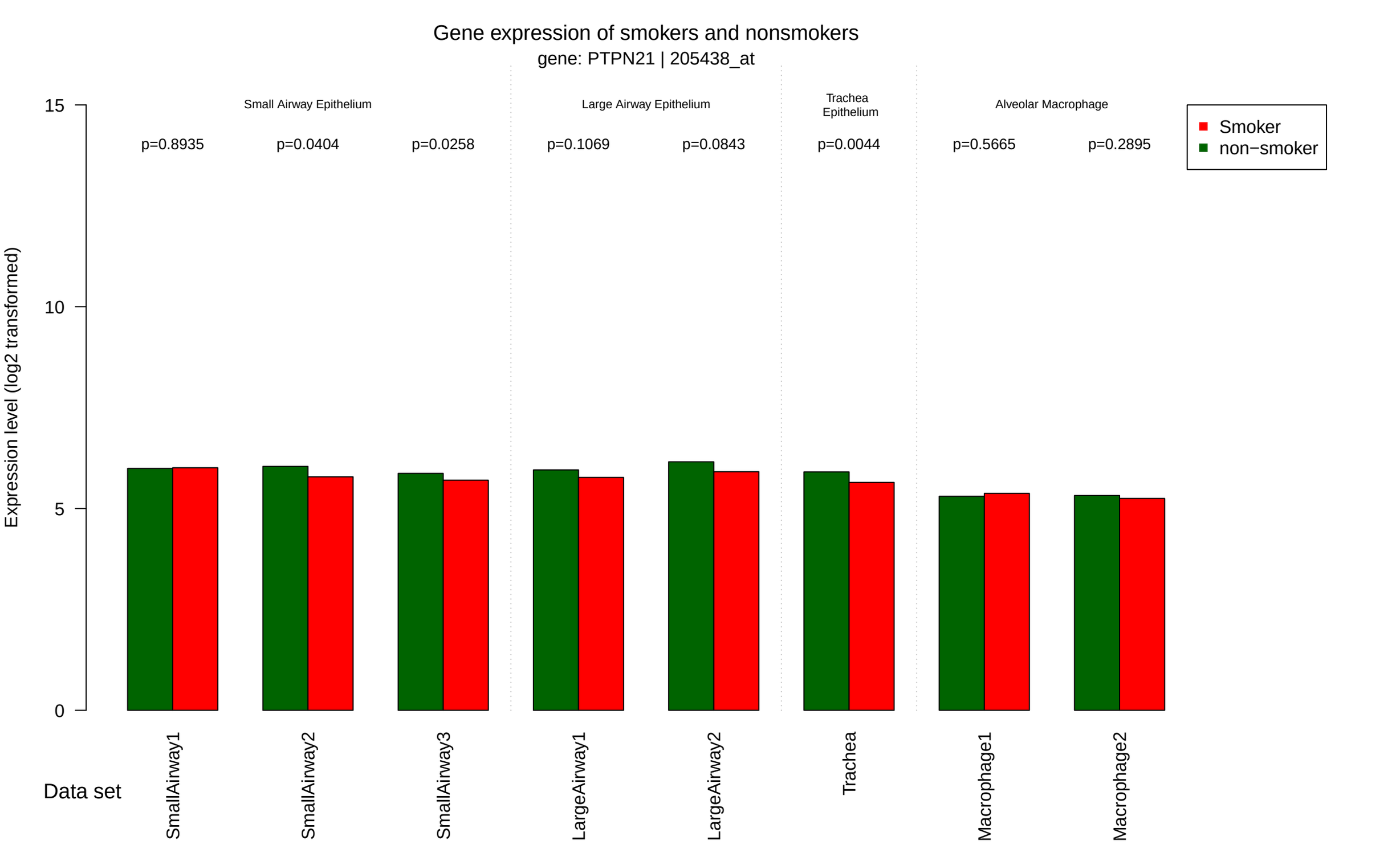
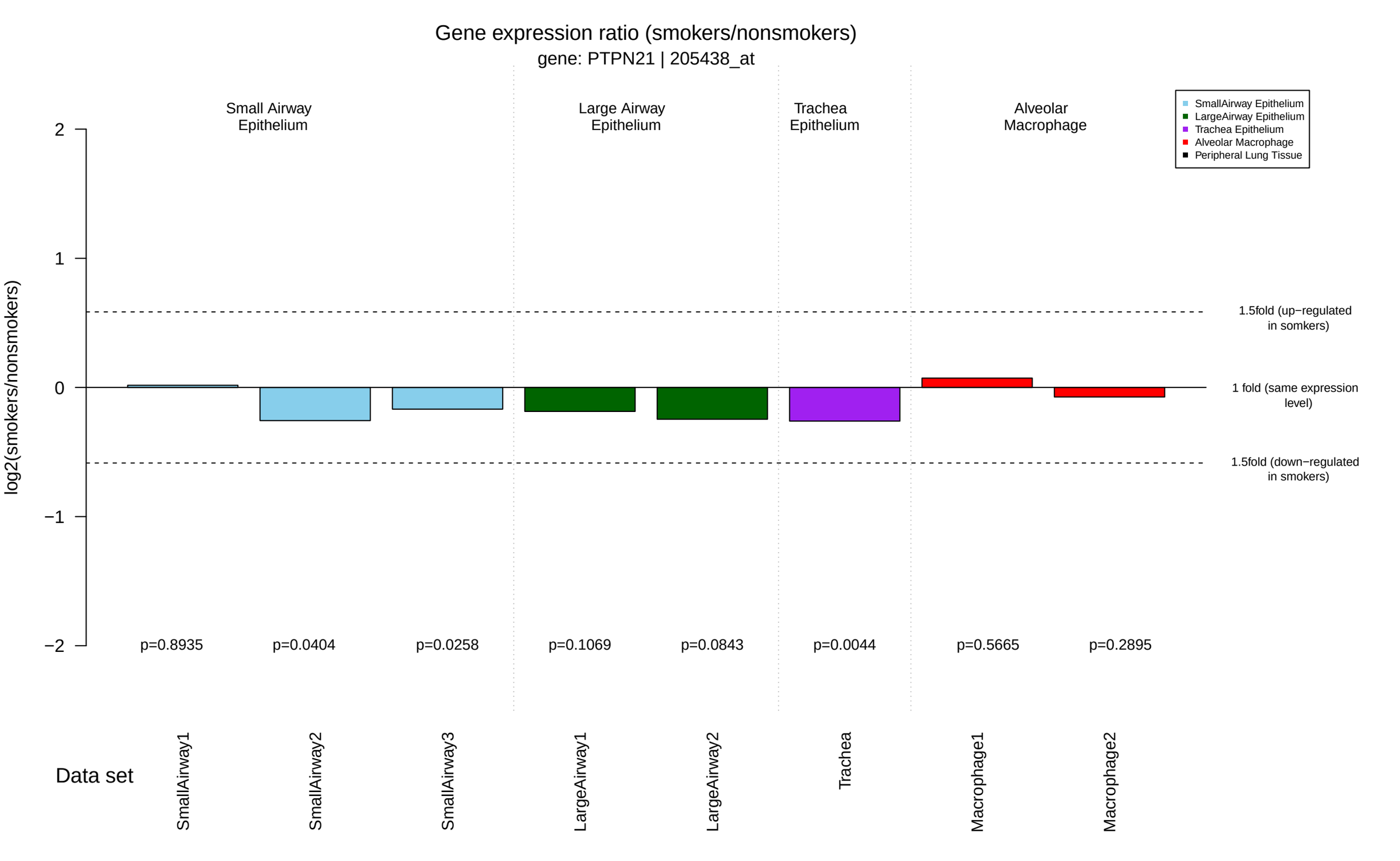
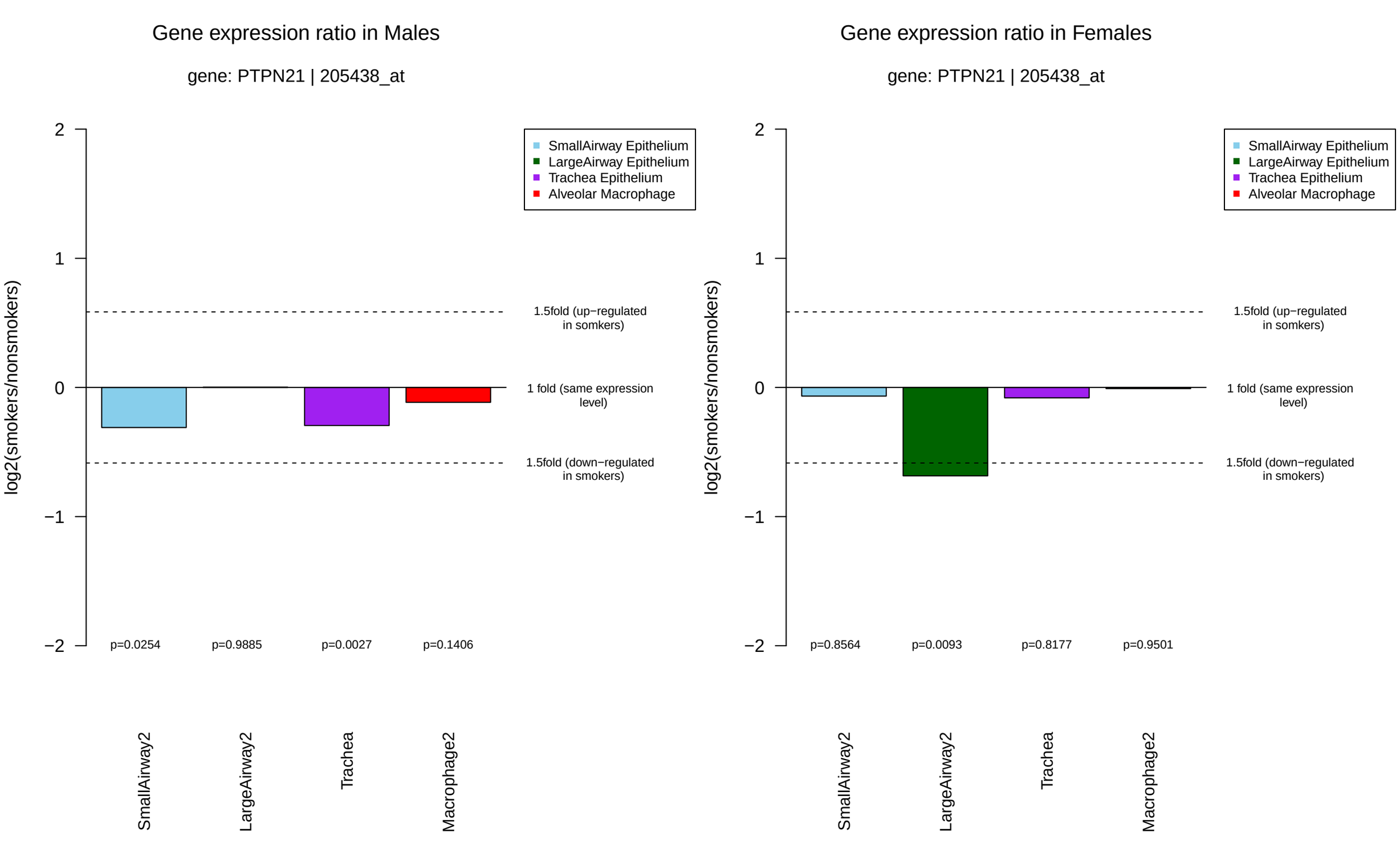
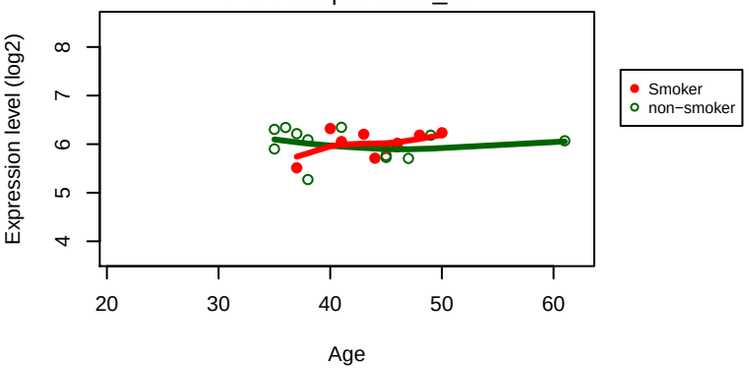
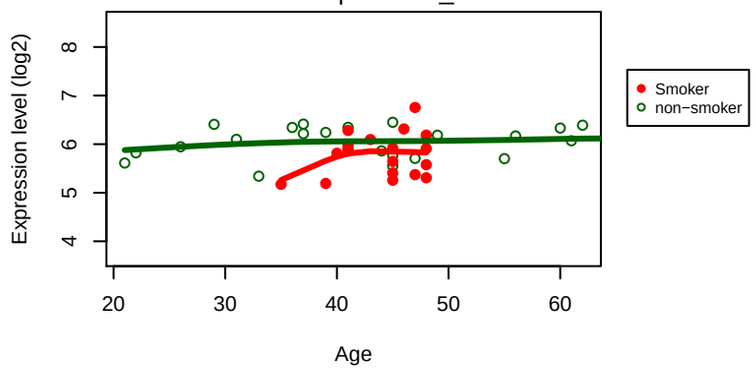
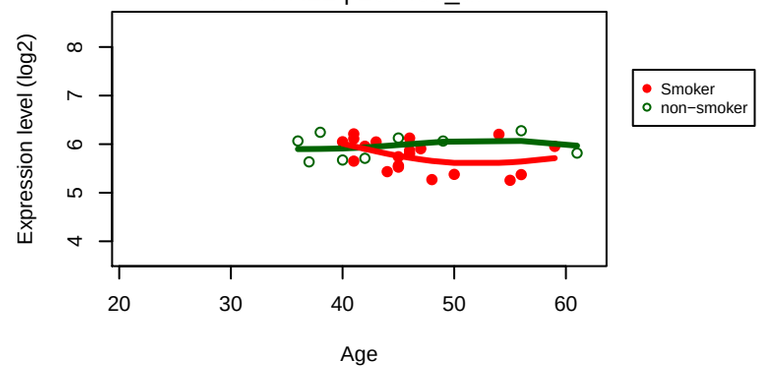

Gene expression of smokers and nonsmokers
gene: PTPN21 | 205438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8935
p=0.0404
p=0.0258
p=0.1069
p=0.0843
p=0.0044
p=0.5665
p=0.2895
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

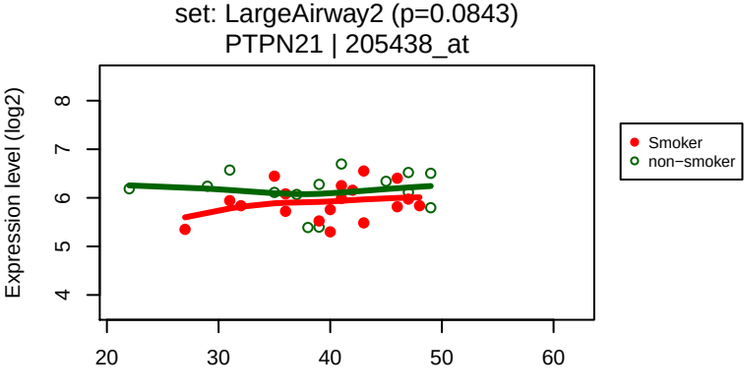

Gene expression ratio (smokers/nonsmokers)
gene: PTPN21 | 205438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8935
p=0.0404
p=0.0258
p=0.1069
p=0.0843
p=0.0044
p=0.5665
p=0.2895
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

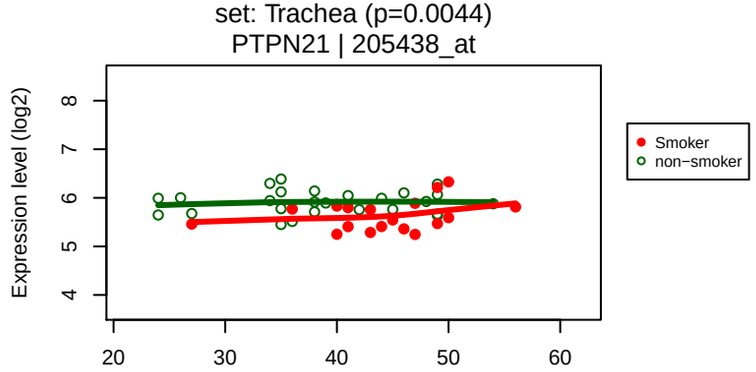

Gene expression ratio in Males
gene: PTPN21 | 205438_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0254
p=0.9885
p=0.0027
p=0.1406
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN21 | 205438_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8564
p=0.0093
p=0.8177
p=0.9501
SmallAirway2
LargeAirway2
Trachea
Macrophage2

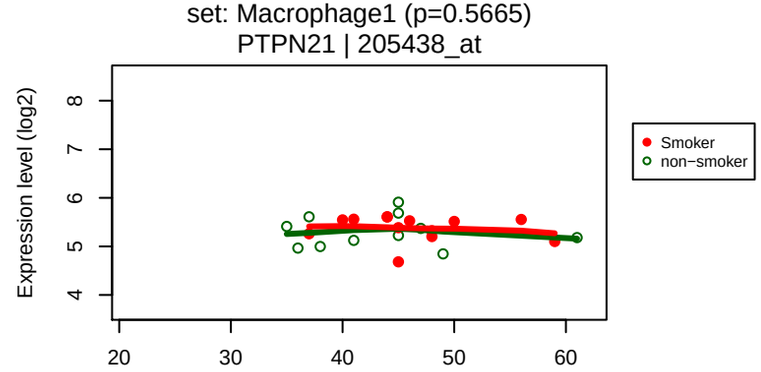

set: SmallAirway1 (p=0.8935)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker

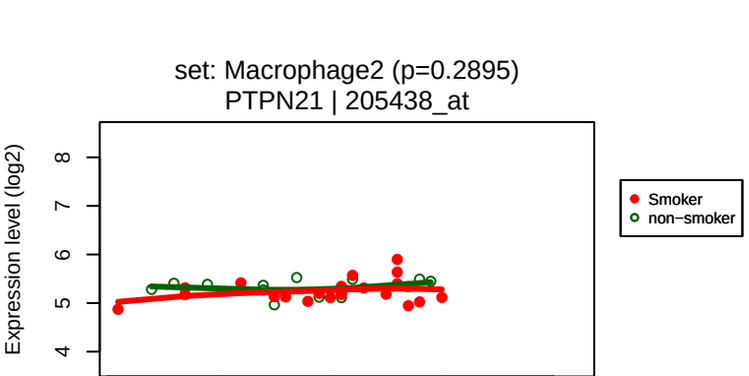

set: SmallAirway2 (p=0.0404)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker

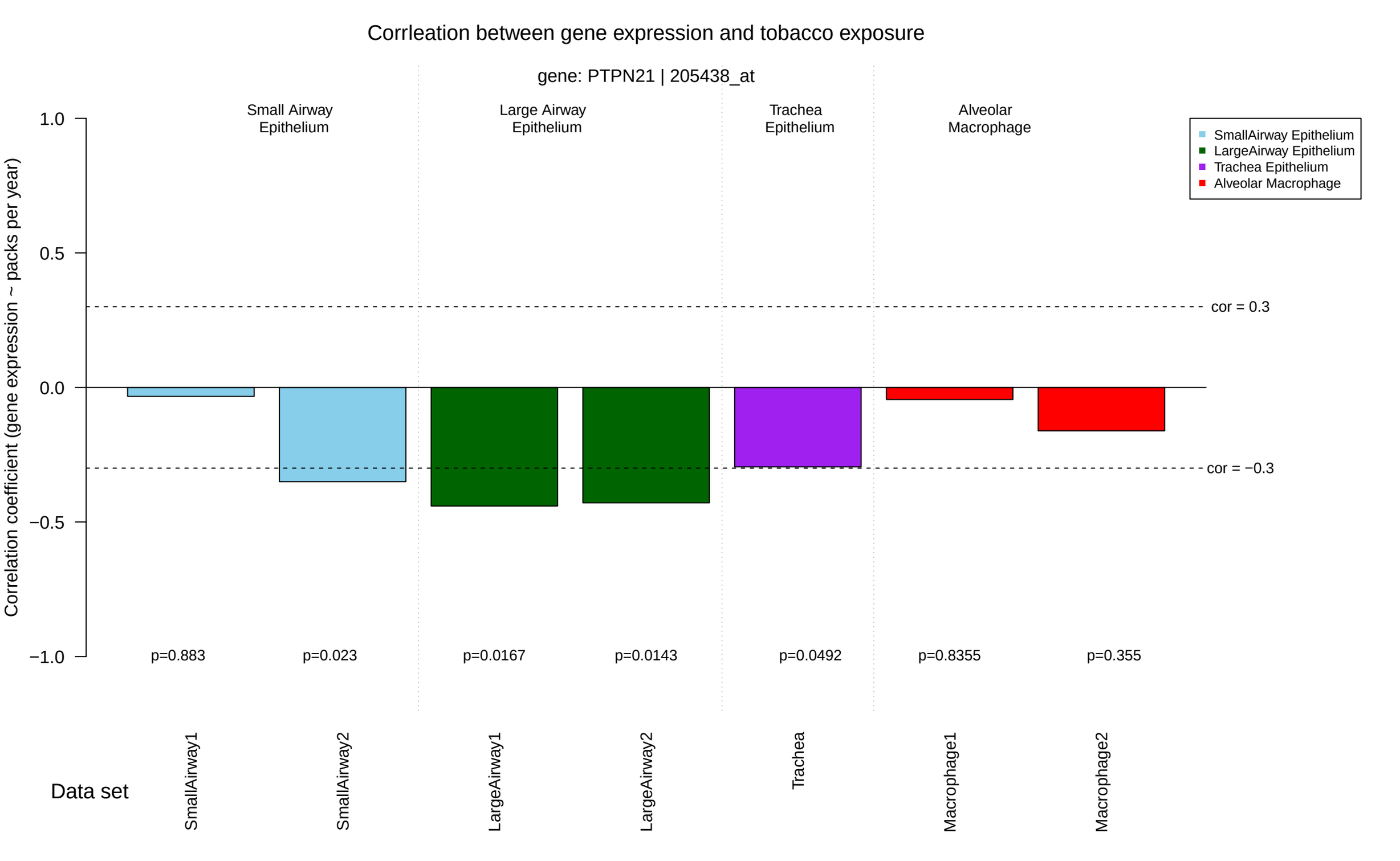

set: LargeAirway1 (p=0.1069)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0843)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0044)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5665)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2895)
PTPN21 | 205438_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN21 | 205438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.883
p=0.023
p=0.0167
p=0.0143
p=0.0492
p=0.8355
p=0.355
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2